

Derek Hissong

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Education

University of Washington, Department of Mechanical Engineering
M.S. Mechanical Engineering

Seattle, WA

Expected: Jan. 2028

University of San Diego, Shiley-Marcos School of Engineering
Dual B.S./B.A. Mechanical Engineering - GPA: 3.6 Overall, 3.8 Major

San Diego, CA

Jan. 2026

Technical Skills

Machine Shop/Lab: 3 & 5-axis CNC, DMA, Microscopy, Lathe, Drill Press, Vertical Mill, Operation Sheet

Computer Software: SolidWorks CAD & CAM, Abaqus, Bluebeam Revu, Excel, ANSYS Fluent, MATLAB

Relevant Coursework: Structural Mechanics, Elasticity, Composite Materials & Material Science, CFD, FEA, Thermofluids, Electrical Circuits, Spanish 1 – 3

Engineering Experience

Composite Engineering Intern

Jun. 2026 - Aug. 2026

Common Fibers

Seattle, WA

- Led intern boat project prototype through CAM, tooling, and lay-up with integrated tool heating elements.
- Manufactured a matrix of composite materials for accelerated UV testing to inform material selection.
- Researched tooling material expansion to inform design constraints of heated tools.
- Assembled cosmetic carbon fiber components for high-end clients such as Hennessy Performance Engineering.

Upcycled Composite Materials Research

Aug. 2025 - Jun. 2026

Shiley-Marcos School of Engineering, University of San Diego

San Diego, CA

- Measured stress, strain, and displacement of composite material samples using Abaqus simulation.
- Operated Dynamic Mechanical Analysis (DMA) machine to collect data on composite sample stress relaxation.
- Validated post-processing results against experimental data from prior testing.
- Presented at the ASME SSDM 2026 conference and documented research results in a report.

Heat Sink & Heat Exchanger CFD

Sep. 2025 - Dec. 2025

Shiley-Marcos School of Engineering, University of San Diego

San Diego, CA

- Used ANSYS Fluent to model fluid dynamics in various environments as a result of changes in heat and airflow.
- Analyzed temperature contours across a heat exchanger resulting from changes in the domain.
- Studied air temperature and velocity contours resulting from heat dissipation from a 3-fin heat sink.
- Refined mesh sizes to reduce interpolation errors and used Fluent Reports to verify model accuracy.

BioBrew Senior Capstone

Sep. 2024 - May 2025

Shiley-Marcos School of Engineering, University of San Diego

San Diego, CA

- Defined project objectives and constraints to enhance team productivity and ensure project success.
- Validated project subsystems using sodium hydroxide (NaOH) gas scrubbing and spectrophotometry.
- Utilized algae sequestration to achieve a 62% average reduction of carbon dioxide (CO₂).
- Benchmarked carbon sequestration effectiveness against OSHA's CO₂ exposure health standards.

Composite Material Boat Hull

Sep. 2024 - Dec. 2024

Shiley-Marcos School of Engineering, University of San Diego

San Diego, CA

- Developed MATLAB code to compute the Classical Laminate Plate Theory (CLPT) of the 15-ply hull lay-up.
- Computed maximal loading scenarios across composite lay-up configurations to optimize laminate design.
- Documented ply analysis and presented to class and professor.

Wobbler Engine

Sep. 2023 - Dec. 2023

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Shiley-Marcos School of Engineering, University of San Diego

San Diego, CA

- Machined components using a lathe, vertical mill, and drill press for wobbler engine construction.
- Created operation sheets with precision standards to optimize manufacturing and engine efficiencies.
- Assembled the highest-efficiency engine in class, operational at 1 psi.

Supplemental Work Experience

Scholar-Athlete Tutor

Sep. 2025 – Dec. 2025

Athletic Department, University of San Diego

San Diego, CA

- Reinforced STEM subject core concepts and problem-solving strategies in individual and group settings.
- Supported scholar-athletes in developing effective study habits, time management, and organization skills.
- Collaborated with athletic department staff to monitor progress towards achieving academic goals.

Project Engineer Intern

Jun. 2023 - Aug. 2023

Rudolph and Sletten General Contractors

San Diego, CA

- Analyzed construction drawings and identified RFI's with Bluebeam Revu and Procore.
- Summarized plan versus field variations using Excel for future cost reconciliation.
- Monitored daily construction activities and communicated with clients and subcontractors.

Assistant Youth Rowing Coach

Jul. 2022 - Aug. 2022

Pocock Rowing Center

Seattle, WA

- Organized and helped lead the daily running of youth rowing camps.
- Helped high school campers develop as rowers and also as holistic athletes and students.
- Provided campers with insight into the transition to collegiate athletics.

Athletics & Leadership

Captain, Division 1 Varsity Men's Rowing

Aug. 2021 - Jul. 2025

University of San Diego - Captain (August 2022 - July 2025)

- Developed exceptional time management through 20+ hours of training per week and a full academic course load.
- Fostered team culture that drove growth from 12 to 45 athletes and improved team GPA from 2.9 to 3.4.
- Established athletic excellence through multiple successes in domestic and international race environments.

FISU World University Rowing Championships

Jun. 2024 - Jul. 2024

USRowing

- Selected for a 4-week training camp, ending with 1 week of international racing in Rotterdam, Netherlands.
- Emphasized workout coordination, cooperative leadership, boat dynamics, and high performance under pressure.

Volunteer Work

Rowers for Change STEM to Stern (StS)

Sep. 2024 - Dec. 2024

University of San Diego

- Spearheaded USD's inaugural involvement with StS to address barriers to rowing and STEM education.
- Collaborated with USD Men's and Women's Rowing programs leadership to increase campaign exposure.
- Engaged alumni and rowing community networks to increase donations and support StS's success.